



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
17-10-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : TEPS, TOPS, I(A), I, I(B), II(A), II & LIVE-I, I(B), II(A)]

Time : 3 Hours

PAPER-1

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS
SECTION-I (i) : (Maximum Marks: 12)

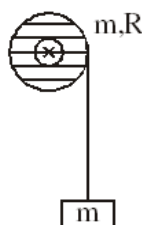
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

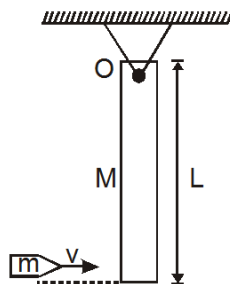
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A block is connected to a disc pulley of same mass 'm', radius R. The acceleration of the block when released.



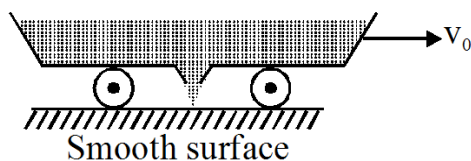
- (A) $\frac{g}{2}$ (B) $\frac{g}{3}$ (C) $\frac{2g}{3}$ (D) g
2. A rod of mass M and length L is suspended by a frictionless hinge at the point O as shown in figure. A bullet of mass m moving with velocity v in a horizontal direction strikes the end of the rod and gets embedded in it. The angular velocity of the rod just after the collision is -



- (A) $\frac{mv}{(3m + M)L}$ (B) $\frac{2mv}{ML}$ (C) $\frac{2mv}{L}$ (D) $\frac{3mv}{(3m + M)L}$

Space for Rough Work

3. A trolley filled with sand moves on a smooth horizontal surface with a velocity 10 m/s. A small hole is made at the base of it from which sand is leaking out at constant rate. As the sand leaks out :-



- (A) The velocity of the trolley increases till 20 m/s.
 (B) The velocity of the trolley decreases till it stops.
 (C) The velocity of the trolley remains 10 m/s.
 (D) The sand applies a horizontal force on trolley towards right while leaking.
4. One end of an ideal spring is fixed at point O and other end is attached to a small disc of mass m which is given an initial velocity v_0 perpendicular to its natural length on a smooth horizontal surface. If the maximum elongation in spring is $\frac{\ell_0}{2}$ then v_0 is (ℓ_0 = natural length and k = stiffness of spring)



- (A) $\frac{\ell_0}{2} \sqrt{\frac{k}{2m}}$ (B) $\frac{3\ell_0}{2} \sqrt{\frac{k}{2m}}$ (C) $\frac{3\ell_0}{2} \sqrt{\frac{k}{5m}}$ (D) $\frac{5\ell_0}{2} \sqrt{\frac{k}{3m}}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

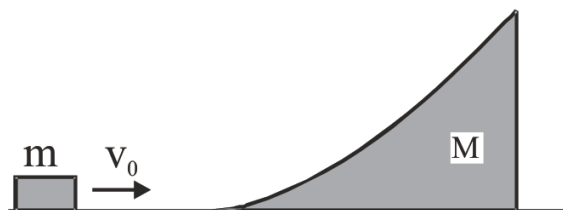
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. A small block of mass m is moving on a horizontal table surface at initial speed v_0 . It then moves smoothly onto a sloped wedge of mass M . The wedge can also move on the table surface. Assume that everything moves without friction. Which of the following statement(s) is (are) correct :-



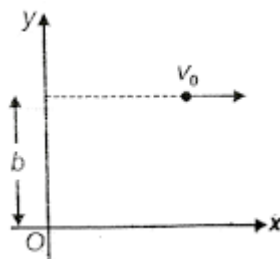
- (A) COM of the system always moves rightward
 (B) COM of the system moves upwards for some time
 (C) Speed of M always increases during contact with m .
 (D) Speed of COM of the system is always constant

Space for Rough Work

6. Choose the correct statements :-

- (A) Whole mass of system is assumed to be concentrated at COM for calculation of moment of inertia
- (B) Whole mass of system is assumed to be concentrated at COM for calculation of gravitational potential energy.
- (C) Whole mass of system is assumed to be concentrated at COM for calculation of linear momentum.
- (D) Whole mass of system is assumed to be concentrated at COM for calculation of kinetic energy.

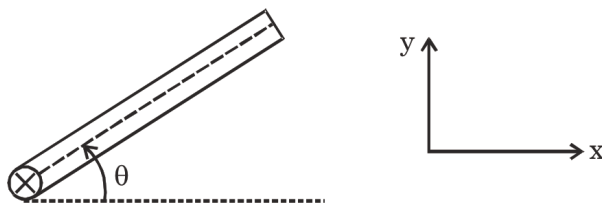
7. A particle of mass m is moving with constant velocity v_0 along the line $y = b$. At time $t = 0$ it was at the point $(0, b)$. At time $t = \frac{b}{v_0}$, the



- (A) Angular momentum of particle about origin is $\sqrt{2} mv_0 b$
- (B) Angular momentum of particle about origin is having value equal to $mv_0 b$.
- (C) Torque acting on particle about origin is zero.
- (D) Torque acting on particle about origin is Non-zero

Space for Rough Work

8. A uniform rod of mass m and length ℓ is hinged at one end which is released from the position shown in the diagram from rest. Rod is free to move in vertical plane, when rod becomes horizontal then :



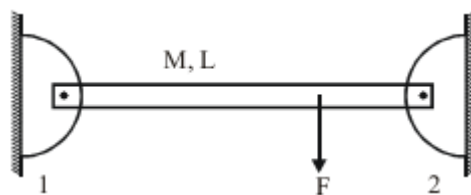
- (A) Acceleration of COM of rod is $-\frac{3g \sin \theta}{2} \hat{i} - \frac{3g}{4} \hat{j}$
- (B) Force applied by hinge on rod is $-\frac{3mg \sin \theta}{2} \hat{i} + \frac{mg}{4} \hat{j}$
- (C) Acceleration of COM of rod is $-\frac{3g \cos \theta}{2} \hat{i} - \frac{g}{4} \hat{j}$
- (D) Force applied by hinge on rod is $-\frac{3mg \cos \theta}{2} \hat{i} - \frac{3mg}{4} \hat{j}$
9. A thin uniform rod of mass m and length ℓ is hanging freely at rest from its topmost point and is free to rotate about its upper end. When it is at rest, it receives an impulse J at its lowest point normal to its length. Immediately after :
- (A) The angular momentum of the rod is $J\ell$
- (B) Angular velocity of the rod is $\frac{3J}{m\ell}$
- (C) The kinetic energy of the rod is $\frac{3J^2}{2m}$
- (D) The linear velocity of the midpoint of the rod is $\frac{3J}{2m}$

Space for Rough Work

10. Choose the incorrect statements :-

- (A) If linear momentum of system is conserved then angular momentum of system must be also conserved.
- (B) If angular momentum conserved then angular velocity must be also conserved about the same axis.
- (C) If net torque about any axis is zero then force must also be zero.
- (D) If net force on a body is zero, torque about an axis through COM may be non-zero.

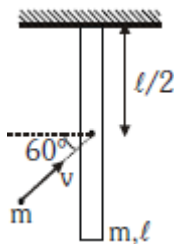
11. A uniform rod of mass M and length L is suspended by two smooth hinges 1 and 2 as shown in the figure. A force F is applied downward at a distance $\frac{L}{3}$ from the hinge 2. Due to the application of force F , the hinge 2 breaks. At this instant, applied force F is also removed. The rod starts to rotate downward about hinge 1. ($g = 10 \text{ m/s}^2$). Choose the correct statement(s) :-



- (A) The reaction at hinge 1, before hinge 2 breaks, is $\frac{F}{3}$.
- (B) The reaction at hinge 2, before hinge 2 breaks, is $\frac{2F}{3}$.
- (C) The reaction at hinge 1, just after breaking of hinge 2 is zero.
- (D) The reaction at hinge 1, just after breaking of hinge 2 is $\frac{F}{6}$.

Space for Rough Work

12. A thin rod of mass m and length ℓ is hinged to a ceiling and it is free to rotate in a vertical plane. A particle of mass m , moving with speed v strikes it as shown in the figure and gets stick with the rod.



- (A) The value of v , for which rod becomes horizontal after collision is $\sqrt{\frac{168}{9}g\ell}$
- (B) The value of v , for which rod becomes horizontal after collision is $\sqrt{\frac{53}{3}g\ell}$
- (C) Angular momentum of (rod + particle) system will remain constant about hinge just before and after collision
- (D) Angular momentum of (rod + particle) system will remain same about centre of mass just before and after collision

Space for Rough Work

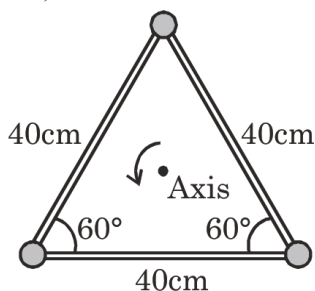
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

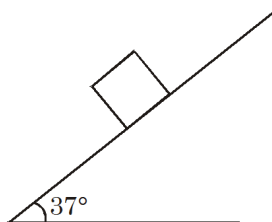
1. The three 200 g masses in figure are connected by massless, rigid rods to form a triangle. What is the triangle's rotational kinetic energy (in joule) if it rotates at 3 rad/s about an axis through the center ?



2. A uniform rod of mass $m = 7\text{ kg}$ and length $l = 1\text{ m}$ is fixed from point A, which is at a distance $l/4$ from one end as shown in the figure. The rod is free to rotate in a vertical plane. The rod is released from horizontal position. Find the reaction force (in N) at the hinge, when kinetic energy of the rod is maximum.

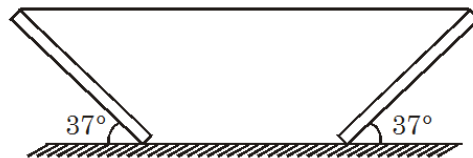


3. A uniform cubical block of side 8m is at rest on an inclined plane. Find distance (in meter) of normal reaction from the centre of block.

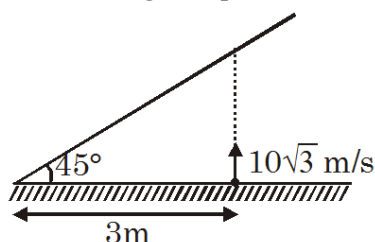


Space for Rough Work

4. Two uniform identical rods of same mass are tied together with the help of a string and balanced as shown in the figure. Find the minimum coefficient of friction for which the system will remain in equilibrium in the position.



5. A bomb is projected vertically upward from ground. At highest point it explodes in two parts having mass ratio 1 : 2. Both the part land on ground at equal distance from point of projection at an interval of 4 sec. If both parts land on ground at time t_1 and t_2 after explosion, find $\frac{t_1 + t_2}{5}$.
6. A particle is thrown upward with speed $10\sqrt{3}$ m/s. It strikes the inclined surface as shown in figure. If collision is perfectly inelastic, find maximum height of particle from ground ($g = 10 \text{ m/s}^2$)



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

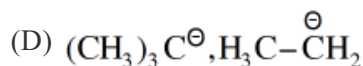
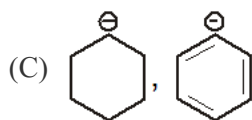
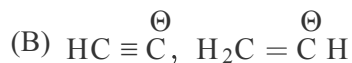
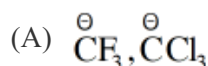
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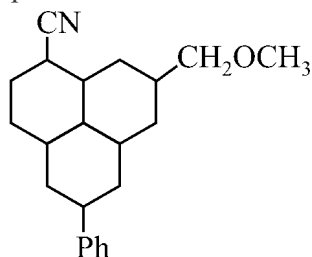
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. In which of the following pairs of carbanion the first one is more stable than second.



2. Degree of unsaturation of following compound is :



- (A) 7 (B) 8 (C) 9 (D) 11
3. Number of moles of electrons present in 6.2 gm of NO_3^- ions is -
- (A) $3.2 \times N_A$ (B) $3.1 \times N_A$ (C) 3.2 (D) 3.1
4. During the complete combustion of methanol, what change in hybridisation does the carbon atom undergo ?
- (A) sp^3 to sp (B) sp^3 to sp^2 (C) sp^2 to sp (D) sp^2 to sp^3

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

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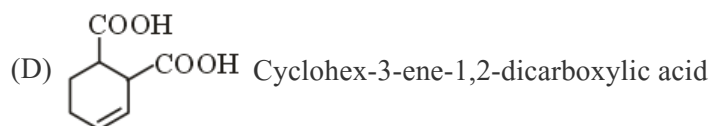
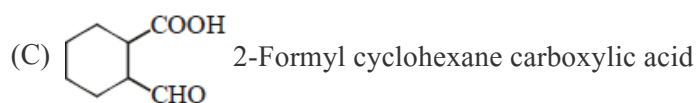
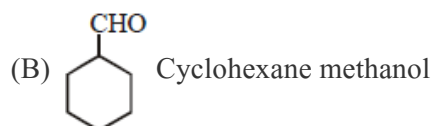
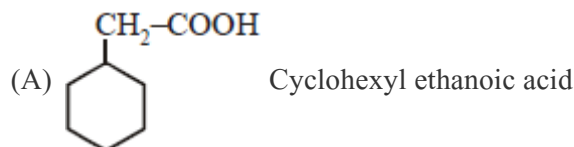
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

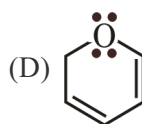
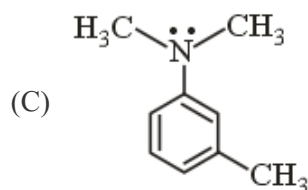
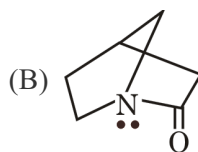
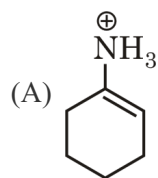
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5. Which of the following IUPAC name are correct matched :



Space for Rough Work

6. In which of the following compound(s) resonance is not present :



7. The number of gram-atom of oxygen in its 8 gram oxygen atoms is equal to number of gram-atom in [At mass of Mg = 24, Fe = 56, Ca = 20, Hg = 200] :

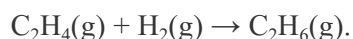
(A) 12 gram Mg

(B) 3.01×10^{22} atoms of He

(C) 20 gram Ca

(D) 5.6 l HCl(g) at 1 atm and 273 K

8. 100 ml of gas mixture of ethene [C_2H_4] and hydrogen which is in excess in mixture was sparked (I), according to the reaction



The volume contracted by 40 ml. Oxygen was then added causing sparking (II) with the final gas mixture and the resulting gas mixture on passing through aq. KOH caused again volume contraction. The correct option(s) is/are (Assume P, T constant)

(A) Volume percentage of ethene in original gas mixture was 60%

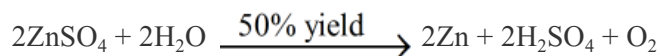
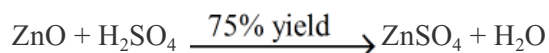
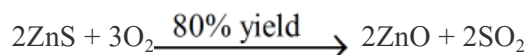
(B) The resulting mixture on passing through KOH caused 80 ml volume contraction

(C) 40 ml of $C_2H_6(g)$ is produced due to first reaction (sparking - I)

(D) Volume percentage of hydrogen gas in original mixture was 60%

Space for Rough Work

9. Given



If 15 mol of ZnS reacts with excess of reactants then which is/are correct statements ?

- (A) 15 mol of Zn is formed
 (B) 2.25 mol of O_2 gas is released in last reaction
 (C) 18 mol of H_2SO_4 is formed in last reaction
 (D) 9 mol of ZnSO_4 is formed in second reaction
10. Which of the following statement(s) is/are correct regarding products formed by the reaction of urea with caustic soda?
- (A) It may be a pyramidal species
 (B) It may be a linear species
 (C) It may have sp^3 hybridisation
 (D) It must have two $\text{p}\pi\text{-p}\pi$ bonds
11. Which of the following orders incorrect between PCl_3F_2 and PCl_2F_3 :-
- (A) $\text{PCl}_3\text{F}_2 > \text{PCl}_2\text{F}_3$ (Dipole moment)
 (B) $\text{PCl}_3\text{F}_2 > \text{PCl}_2\text{F}_3$ (Total number of lone pair)
 (C) $\text{PCl}_3\text{F}_2 > \text{PCl}_2\text{F}_3$ (Cl-P-Cl Bond angle)
 (D) $\text{PCl}_3\text{F}_2 > \text{PCl}_2\text{F}_3$ (P-Cl Bond length)
12. Choose the CORRECT order of their mentioned property.
- (A) $\text{NH}_3 < \text{PH}_3 < \text{SbH}_3$: E-H bond length
 (B) $\text{O}^+ > \text{N}^+$: IE order
 (C) $\text{Cl} > \text{B}$: E.N. order
 (D) $\text{Rb} < \text{Li}$: Reducing nature

Space for Rough Work

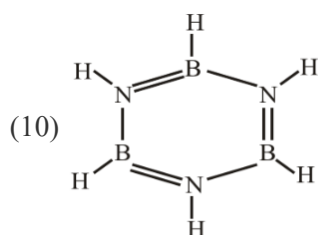
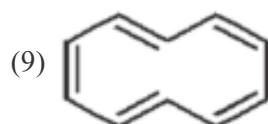
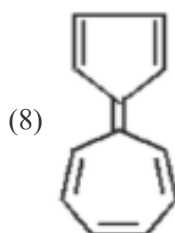
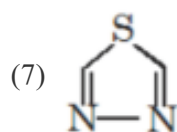
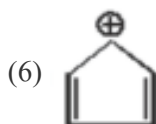
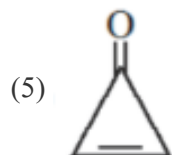
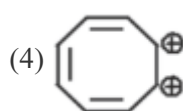
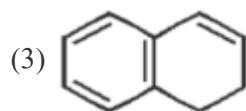
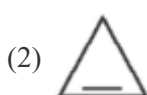
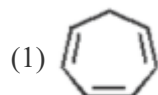
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. How many of following compound are aromatic in nature



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2. Number of carbanions which are more stable than $\overset{\ominus}{\text{C}}\text{H}_2 - \text{CH}_3$
- (i) $\overset{\ominus}{\text{C}}\text{H}_3$ (ii) $\overset{\ominus}{\text{C}}\text{F}_3$ (iii) $\overset{\ominus}{\text{C}}\text{H}(\text{CH}_3)_2$ (iv) $\overset{\ominus}{\text{C}}\text{Cl}_3$
 (v) $\overset{\ominus}{\text{C}}\text{H}_2 - \text{CCl}_3$ (vi) $\overset{\ominus}{\text{C}}\text{H}_2 - \text{CH}_2 - \text{CH}_3$ (vii) $\overset{\ominus}{\text{C}}\text{H}_2 - \overset{\oplus}{\text{N}}(\text{CH}_3)_3$ (viii) $\overset{\ominus}{\text{C}}(\text{CH}_3)_3$
3. For the given redox reaction,
 $x\text{KMnO}_4 + y\text{H}_2\text{O}_2 + z\text{H}_2\text{SO}_4 \rightarrow p\text{MnSO}_4 + 10\text{O}_{2(\text{g})} + q\text{K}_2\text{SO}_4 + r\text{H}_2\text{O}$
 Calculate value of $(x + z - p)$
4. The density of a gas at 27°C and 1 atm pressure is d . Pressure remaining constant, the temperature (in kelvin) at which its density is $0.5d$ is :-
5. In how many of the following species % s - character of all bonds is not identical.
 IF_7 , CH_4 , SF_4 , XeF_3^+ , OCl_2 , ICl_2^- , BF_3 , ClOH , 
6. Find the number of species from given below which have Tetrahedral unit(s) :
- | | |
|--------------------------------------|-------------------------------------|
| $\text{N}_2\text{O}_{5(\text{s})}$ | $\text{PCl}_{5(\text{s})}$ |
| $\text{PBr}_{5(\text{s})}$ | $\text{NH}_4\text{Cl}_{(\text{s})}$ |
| $\text{Na}_2\text{SO}_{4(\text{s})}$ | $\text{XeF}_{6(\text{s})}$ |
| $\text{H}_4\text{P}_2\text{O}_7$ | Pyrosilicate |

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PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If A(8, 4), B(20, 10), C(11, 1) are vertices of ΔABC , then distance between orthocentre and circumcentre is -
 (A) $5\sqrt{3}$ (B) $3\sqrt{5}$ (C) $4\sqrt{3}$ (D) $2\sqrt{5}$
2. In triangle ABC, AB = 13, BC = 14 and CA = 15. A circle of radius r passes through point A and is tangent to line BC at C, then r is equal to -
 (A) $\frac{75}{8}$ (B) $\frac{25}{2}$ (C) $\frac{75}{4}$ (D) $\frac{25}{4}$
3. In ΔABC , $\angle A$, $\angle B$ & $\angle C$ forms an A.P.. The length of altitude from B to AC is equal to $\ell(AB) - \ell(BC)$. The value of $\sin\left(\frac{C - A}{2}\right)$ is - (where $\ell(AB)$ denotes length of AB)
 (A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) 0 (D) 1
4. Locus of the centroid of the triangle whose vertices are $(a \cos\theta, a \sin\theta)$, $(b \sin\theta, -b \cos\theta)$ and $(0,1)$, where θ is a parameter, is
 (A) $(3x - 1)^2 + (3y)^2 = a^2 + b^2$ (B) $(3x)^2 + (3y - 1)^2 = a^2 + b^2$
 (C) $(3x - 1)^2 - (3y)^2 = a^2 - b^2$ (D) $(3x)^2 - (3y - 1)^2 = a^2 - b^2$

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SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

-
5. All the roots of equation $x^3 + ax^2 + bx + c = 0$ are positive integers and are greater than 2 where $a, b, c, \in \mathbb{R}$. If $a + b + c + 2010 = 0$, then a cannot be
- (A) -52 (B) -56 (C) -58 (D) -64
6. If the variable line $L \equiv y - mx - c = 0$ ($m, c \in \mathbb{R}$ and $c \neq 0$) cuts the curve $y = x^2 - x$ at P and Q such that $\angle POQ = \frac{\pi}{2}$ where O is origin, then
- (A) $m + c - 1 = 0$ (B) $m - c + 1 = 0$
- (C) line L always passes through (1, 1) (D) line L always passes through (-1, -1)
-

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7. Let A(2, 0) and B(0, 2) be 2 points then coordinates of point P on $2x + 3y + 1 = 0$ will be -
- (A) (7, -5) if $|PA - PB|$ is maximum (B) $\left(-\frac{1}{5}, -\frac{1}{5}\right)$ if $|PA - PB|$ is maximum
- (C) (7, -5) if $|PA - PB|$ is minimum (D) $\left(-\frac{1}{5}, -\frac{1}{5}\right)$ if $|PA - PB|$ is minimum
8. Let $S_1, S_2, \dots$ be squares such that for each $n \geq 1$, the length of a side of S_n equals to the length of a diagonal of S_{n+1} . If the length of a side of S_1 is 10 cm, then for which of the following value (s) of n is the area of S_n less than 1 sq. cm ?
- (A) 7 (B) 8 (C) 9 (D) 10
9. The equation of the straight line equally inclined to the coordinate axes and equidistant from the points (2,0) and (0,4) can be-
- (A) $y = x - 1$ (B) $y = x + 1$ (C) $x + y = 3$ (D) $x + y + 3 = 0$

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10. In $\triangle ABC$, equation of side BC is $x + y = 6$, the circumcentre and orthocentre are (3,1) and (2,2) respectively. Then
- (A) reflection of orthocentre in line BC is (4,4) (B) circumradius of $\triangle ABC$ is $\sqrt{10}$
- (C) circumradius of $\triangle ABC$ is $\frac{\sqrt{10}}{2}$ (D) centroid of $\triangle ABC$ is $\left(\frac{8}{3}, \frac{4}{3}\right)$
11. A straight rod of length 9 units slides with its ends A and B being always on the X and Y axes respectively. Then the locus of the centroid of $\triangle OAB$ is not equal to
- (A) $x^2 + y^2 = 3$ (B) $x^2 + y^2 = 9$ (C) $x^2 + y^2 = 1$ (D) $x^2 + y^2 = 81$
12. The largest common element of two sets $\{1,4,7,10,\dots,541\}$ and $\{3,8,13,18,\dots,498\}$ is less than
- (A) 493 (B) 495 (C) 496 (D) 497

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SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Consider Rhombus ABCD with vertex A(3,1) and whose one diagonal is $3x - 4y + 5 = 0$. If the equation of line through mid points of sides AB and AD is $ax + by + c = 0$ where a,b,c are integers then minimum value of $|a + b + c|$ is
2. The area bounded by the curves $y \geq \sqrt{3}x$, $y \geq -\sqrt{3}x$ and $0 \leq y \leq 2\sqrt{3}$ is $p\sqrt{q}$ then $(p + q)$ is (p and q are coprime)
3. Sum of first three terms of G. P. with non-integer common ratio is 21 and sum of their squares is 189. If S denotes sum of infinite term of this series, then $\frac{S}{4}$ is equal to
4. The number of solution(s) of trigonometric equation $4\cos^2x + \tan^2x + \cot^2x + \sec^2x = 6$ in $[0, 2\pi]$ is (are)
5. If α, β, γ are the roots $x^3 + x^2 - 2x + 1 = 0$ and $\Delta = \begin{vmatrix} \gamma^2 & \beta^2 & 2\beta\gamma - \alpha^2 \\ 2\alpha\gamma - \beta^2 & \alpha^2 & \gamma^2 \\ \alpha^2 & 2\alpha\beta - \gamma^2 & \beta^2 \end{vmatrix}$, then $\sqrt{\Delta}$ is
6. The distance of the point (1,2) from the line $y = 2x + 3$ measured along the line $y = x$, is

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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